Adrian Sizemore

Principal Network Engineer | Apex, NC | adriansizemore@gmail.com

Professional Summary

Adrian is a technical leader who evaluates emerging technologies, translates complex infrastructure risks into business decisions, and guides cross-functional teams through large-scale modernization. His experience spans carrier-grade MPLS backbones, global WAN and circuit architecture, Cisco ACI and Arista data-center fabrics, secure network segmentation, deep incident analysis, and Python/Django/React automation. He combines hands-on engineering depth with vendor management, financial delivery, standards development, and executive technical advising.

Professional Experience

Principal Network Engineer | MetLife

May 2022 - Present | Cary, NC

Leads global infrastructure strategy, modernization, transport architecture, and Tier-3 technical consulting for high-availability environments. Advises leadership while remaining the final escalation point for complex outages, network redesigns, and operational governance.

Strategic Leadership and Financial Management

Partners with senior executives and procurement teams to align infrastructure strategy with financial delivery, budget processes, depreciation schedules, operational run-rates, and portfolio outcomes.

Global Expert Consulting and Tier-3 Operations

Provides deep technical consulting for international data centers, leads forensic analysis for high-visibility outages, and works directly with provider Tier-3 engineers to isolate complex transport failures.

Technology Ownership and Modernization

Led evaluation, vendor scoring, architecture, and adoption of Arista EOS as a replacement for legacy Cisco ACI environments experiencing chronic stability issues.

Infrastructure Strategy and Future-Proofing

Defined modernization standards including single-mode fiber for 400G+ readiness, unified hardware strategies, and deep observability across the infrastructure estate.

Global Carrier Strategy and Transport Engineering

Architects high-availability WAN and data-center interconnect solutions across dark fiber, waves, and MPLS while enforcing provider SLA accountability and validating carrier designs.

Innovation and Automation

Architected React and Django network applications and a high-performance validation engine capable of auditing more than 10,000 IP addresses in seconds.

Operational Governance

Serves as a final technical escalation point, approves global Methods of Procedure and network designs, and leads root-cause analysis that drives systemic redesign.

- Engineered a pre-change and post-change validation engine that audits more than 10,000 IP addresses in seconds and stores results for historical analysis.
- Led evaluation and architecture for replacing legacy Cisco ACI environments with Arista EOS to address stability and lifecycle requirements.

- Established single-mode fiber as a modernization standard to support 400G and higher-speed infrastructure readiness.
- Led deep-dive forensic analysis and root-cause resolution for high-visibility global network outages and architectural failures.

Lead Network Engineer | MetLife

Oct 2017 - May 2022 | Cary, NC

Designed secure Cisco ACI fabrics and served as the primary escalation SME for complex BGP/MPLS and production incidents. Built automation and validation tooling that reduced manual work and improved change safety.

Secure Fabric Design

Designed and deployed large-scale Cisco ACI fabrics, complex segmentation policy, Endpoint Groups, endpoint reachability, and fabric-wide connectivity controls.

L4-L7 Integration

Designed, deployed, and troubleshot Citrix NetScaler load-balancing solutions integrated with Cisco ACI fabrics.

Automation and Intent-Based Validation

Developed Python and YAML workflows, configuration middleware, and a web-based tool for validating network changes before and after deployment.

SME and Escalation Leadership

Led high-impact outages and served as the escalation SME for complex Layer-3 BGP/MPLS backbone problems.

Principal Backbone Engineer | Verizon

Jan 2016 - Oct 2017

Led Tier-3 engineering for a nationwide carrier-grade MPLS backbone. Diagnosed BGP peering anomalies, managed large-scale traffic engineering events, and developed proactive reliability tooling.

Carrier-Grade Operations

Led critical Tier-3 backbone operations, BGP troubleshooting, and nationwide MPLS traffic engineering.

Proactive Reliability

Developed automated fault-detection and performance-monitoring scripts intended to reduce customer impact and Mean Time to Repair.

Network Engineer (Tier 1 / Tier 2) | Verizon

Jan 2005 - Jan 2016

Built extensive hands-on experience across physical-layer troubleshooting, circuit provisioning, data-center and WAN operations, and Cisco hardware lifecycles. Supported mission-critical infrastructure upgrades and carrier operations.

Projects

Network Change Validation Engine

Network automation

A full-stack pre-change and post-change validation platform that tests network reachability at scale. It audits more than 10,000 IP addresses in seconds and retains results for historical comparison.

Problem

Manual validation was too slow and inconsistent for large network changes and did not provide durable historical evidence.

Approach

Built a Python and Django REST Framework backend with a React interface to execute high-volume validation and preserve network-state results.

Outcome

Replaced manual checks with fast, repeatable validation capable of processing more than 10,000 IP addresses in seconds.

Routing Policy Analysis Tool

Network analysis

A routing-analysis tool that performs longest-prefix matching for an IP address or network segment. It identifies the routing policies that can affect the requested destination.

Problem

Engineers needed a faster and more reliable way to determine which routing policies applied to specific addresses and prefixes.

Approach

Implemented longest-prefix analysis and policy correlation in a purpose-built network tool.

Outcome

Reduced the manual analysis required to trace routing-policy impact.

Intent-Based Network Automation Framework

Configuration automation

A YAML-driven framework that converts engineering intent into repeatable network configurations. It supports retrieval, modification, generation, validation, and deployment of Cisco configuration at scale.

Problem

Large-scale configuration work depended on repetitive manual processes that were difficult to validate consistently.

Approach

Combined Python, YAML, Jinja2, APIs, and configuration workflows to generate and apply network changes from structured intent.

Outcome

Improved configuration consistency and provided an extensible automation pattern for internal engineering teams.

Load Balancing Request Validation

API automation

An API-driven workflow for validating load-balancing requests and tracking network state. It applies repeatable checks to reduce errors in L4-L7 service delivery.

Problem

Load-balancing requests and configuration changes required consistent technical validation and state tracking.

Approach

Designed API-based validation around Citrix NetScaler and data-center network requirements.

Outcome

Created a repeatable validation path for load-balancer service requests.

Skills and Technologies

Enterprise Architecture

Global WAN Design, Multi-Carrier Circuit Engineering, Data Center Interconnect, Spine-Leaf Fabrics, Disaster Recovery Strategy, Single-Mode Fiber Architecture, Capacity Planning, High-Level Network Design, Low-Level Network Design

Networking and Infrastructure

Arista EOS, Cisco ACI, Cisco IOS, Cisco IOS XR, Cisco NX-OS, Cisco Nexus, BGP, OSPF, EIGRP, MPLS, VXLAN, EVPN, QoS, Spanning Tree, vPC, NAT, DNS, Infoblox, SD-WAN, Multicast, Dark Fiber, Optical Waves, Ciena Optical Transport, Citrix NetScaler, Palo Alto Firewalls

Network Automation and Software

Python, Django, Django REST Framework, React, JavaScript, TypeScript, REST APIs, YAML, JSON, Jinja2, Git, Linux, Docker, Kubernetes

Security and Observability

Network Segmentation, DDoS Mitigation, Secure Fabric Design, Packet Capture, Wireshark, Flow Analysis, NetFlow, Network Telemetry, Network State Validation, Root Cause Analysis

Operations and Leadership

Executive Technical Advising, Incident Response Leadership, Vendor Management, RFP Management, Contract Negotiation, Procurement Planning, Budgetary Forecasting, Financial Asset Management, Technical Documentation, Methods of Procedure, Cross-Team Collaboration, Agile, Scrum, SLA Management

Military Service

U.S. Navy | Information Technician

Service dates are documented as 1999-2003; exact dates were not present in the source resumes.

U.S. Marine Corps | Field Radio Operator

Service dates are documented as 1992-1996; exact dates were not present in the source resumes.

Education

East Carolina University [in progress]

Bachelor of Science in Industrial Technology | Information Systems and Cybersecurity

Currently enrolled

Completion and attendance dates were not present in the source resumes.

Wake Technical Community College

Associate of Applied Science | Telecommunications and Network Engineering Technologies

Completed

Completion and attendance dates were not present in the source resumes.

Certifications

Cisco Certified Internetwork Expert (CCIE), Routing and Switching

Cisco | Active | 51766

Cisco Certified Network Professional (CCNP)

Cisco | Past

Cisco Certified Internetwork Professional (CCIP)

Cisco | Past